[1]

① True.

Suppose f is measurable but f is not constant. Then there exists $C_1 \neq C_2$, s.t. $\{f=C_1\} \neq \emptyset$ and $\{f=C_2\} \neq \emptyset$.
However, $\{f=C_1\} \neq \Omega$, so $\{f=C_1\} \notin \mathcal{A}$, contradicting measurability of f. So f must be constant.

② False.

Suppose f is not measurable and $g=-f$. So neither f or g is measurable, but $f+g=0$

③ True

f is measurable, so f^2 is also measurable. Then $\{f^2 \leq g\} \in \mathcal{A}$

④ False.

Suppose set B is not measurable, define $f(x)=\begin{cases} 1 & x \in B \\ -1 & x \in B^c \end{cases}$
then $f^2=1$, which is measurable. But $\{f=1\}=B \notin \mathcal{A}$,
So f is not measurable.

⑤ True. It can be obtained by Lemma 1.2.2, since $f^3=g(f)$,
where $g(x)=x^{\frac{1}{3}} : (\bar{R}, \mathcal{B}(\bar{R})) \to (\bar{R}, \mathcal{B}(\bar{R}))$ is measura

⑥ False. Suppose f is a constant function such that $f(\omega)=C$, where C is a constant. Then $f(B)=C$ for $B \in \mathcal{A}$ with $B \neq \emptyset$.

⑦ True. Since g is measurable, $1\{g>0\}$ is also measurable, so is $f \cdot 1\{g>0\}$ by Theorem 1.3.13.

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⑧ False. Consider case ①. Let $A \subset \Omega$ with $A \neq \emptyset$ and $A \neq \Omega$

Define $f = \begin{cases} \infty & \text{on } A \\ -\infty & \text{on } A^c \end{cases}$. Then $\{f > 0\} = A \neq A$.

But $f \cdot 1_{\{|f| \leq n\}} = 0$, which is measurable.

⑨ False. Consider measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$

Note that $\mathcal{B}(\mathbb{R})$ is smaller than the power set of $\mathbb{R}$

i.e. $\exists A \subset \mathbb{R}$, but $A \notin \mathcal{B}(\mathbb{R})$.

If $\alpha \in A$, define $f_\alpha(x) = \begin{cases} 1 & x = \alpha \\ 0 & x \neq \alpha \end{cases}$.

If $\alpha \notin A$, define $f_\alpha(x) = 0$. Then $\sup_{\alpha \in \mathbb{R}} f_\alpha(x) = 1_A$,

which is not measurable.

[2] $h(w) = f(w) \cdot 1_A + g(w) \cdot 1_{A^c}$. Since $f(w) \cdot 1_A$ and $g(w) \cdot 1_A$ are all measurable functions, $h(w)$ is also measurable.

[3] Suppose $f(x) = \sum_{j=1}^{n} a_j 1_{A_j}$ is a simple function, where $A_i \cap A_j = \emptyset$ for $i \neq j$, $\bigcup_{j=1}^{n} A_j = \Omega$.

So $f(x) = \sum_{\{j : a_j \geq 0\}} a_j 1_{A_j} + \sum_{\{j : a_j < 0\}} a_j 1_{A_j}$ and we have

$f^+ = \sum_{\{j : a_j \geq 0\}} a_j 1_{A_j}$ $f^- = \sum_{\{j : a_j < 0\}} a_j 1_{A_j}$, so $f^+, f^- \in \mathcal{E}$.

If $f^{\pm} \in \mathcal{E}$, then $f = f^+ - f^-$ is measurable and takes only finite different values, so $f \in \mathcal{E}$.

[4] Consider an $w \in \Omega$. If $u(w) \geq 0$, $u^+(w) = u(w)$ and $u^-(w) = 0$, hence $u(w) = u^+(w) - u^-(w)$. If $u(w) < 0$, $u^+(w) = 0$, $u^-(w) = -u(w)$, so $u(w) = u^+(w) - u^-(w)$. With

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